CS & DA
Probability and Statistics

Probability

Lecture No. 04



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Recap of previous lecture



Topic

Probability (Part 3)

Topics to be Covered



Topic

BASICS of PROBABILITY (Part-04)

(Conditional Probability)

Conceptual Questions of Conditional Prob

(just Reduce the s.space according to condition)

Qs A Couple has 3 kids, If 1st child is known to be a Boy then Find the prob that there are exactly 2B in the family.

Condition

Sol: original S.Sp = {(BBB), (BBG), (BGB), (BGG), (GBB), (GBG), (GGB), (GGG)} = 8

After I

Reduced S.Sp = {1st child is Boy} = {(BBB), (BBG), (BGB), (BGG)} = 4

(Acc to Condition)

fav Cases = {exactly 2B} = {(BBG), (BGB)} = 2

$$\text{Conditional Prob} = \frac{\text{fav Cases}}{\text{R-S.Sp}} = \frac{2}{4} = \frac{1}{2} \neq \frac{4}{8}$$

App II : Req Prob $= P\left[\frac{\text{exactly 2 B}}{1^{st}\text{ child is Boy}}\right] = P[(\textcircled{B}BG) \text{ or } (\textcircled{B}GB)]$ ← ME →

$= 1\times\frac{1}{2}\times\frac{1}{2} + 1\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{2}$

Note: A Couple has 3 kids then find the prob that there are exactly 2 B in a family?

Sol: Req Prob $= \frac{\text{fav}}{\text{Total}} = \frac{{}^3C_2}{2^3} = \frac{3}{8}$

& with Cond^n Ans $= \frac{2}{4}$

Q. two Int are to be selected from integers 1, 2, 3, 4 ..... 11.
If their Sum is Even then Find the prob that Both the selected Integers are odd?

Condition

App I — Lengthy — Do yourself.

App II — 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

$^{11}C_2 = 55$ pair

2, 4, 6, 8, 10 → $^5C_2 = 10$ pair

1, 3, 5, 7, 9, 11 → $^6C_2 = 15$ pair

Total ways of selecting 2 int = $^{11}C_2 = 55$ pair

Reduced ways : : : : : = {Sum is Even}

= Either (Both are odd) or (Both are Even)

= $^6C_2 + ^5C_2 = 15 + 10 = 25$ Pair

fav Cases = {Both are odd} = $^6C_2 = 15$

req Prob = $\frac{\text{fav Cases}}{\text{R. Cases}} = \frac{15}{25} = \frac{3}{5}$

Analysis: ① With Cond^n, Req Prob $= P\left(\frac{\text{Both are odd}}{\text{Sum is Even}}\right) = \frac{{}^6C_2}{{}^6C_2 + {}^5C_2} = \frac{15}{15+10} = \frac{15}{25}$

② Had the Cond^n were not these then ans would have been $= ? = \frac{{}^6C_2}{{}^{11}C_2} = \frac{15}{55}$

HW8 App I :- Do yourself.

$S = \{(1,1)(1,2)(1,3)(1,4)\ldots(1,10)(1,11)$
$(2,1)(2,2)(2,3)(2,4)\ldots(2,10),(2,11)$
$(3,4),(3,5)\ldots(3,11)\ldots(10,11)\}$

$= {}^{11}C_2 = 55$ pair

Alpha (RMA) S.space = {(1,2), (1,3), (1,4) ..... (1,10), (1,11), (2,3), (2,4), (2,5) ..... (2,10), (2,11), (3,4), (3,5) ..... (3,11), ... (9,10), (9,11), (10,11)} = 55 pair

Reduced S.Sp (Sum is Even) = {Both odd: (1,3), (1,5) ... (1,11), (3,5), (3,7), (3,9), (3,11) or (5,7), (5,9), (5,11) ..... (9,11) → 15; Both Even: (2,4), (2,6), (2,8), (2,10), (4,6), (4,8), (4,10) ..... (8,10) → 10} + i.e 25 pair

Fav Cases = {Both are odd} = {(1,3), (1,5) ... (1,11), (3,5) ... (9,11)} — 15 pair

Conditional Prob = $\frac{\text{Fav Cases}}{\text{RS.Sp}} = \frac{15}{25}$

Q. There are two Gangsters MUNNA MOBILE (= A) & PAPPU PAZER (= B). They both fire at the Target once with prob of their hitting is $\frac{4}{5}$ & $\frac{3}{4}$ resp. then find the prob,

$P(A) = \frac{4}{5}$, $P(B) = \frac{3}{4}$, A & B are Independent.

$P(\bar{A}) = \frac{1}{5}$, $P(\bar{B}) = \frac{1}{4}$

App III (1) Both will hit = ? $\Rightarrow P(A \cap B) = P(A) \cdot P(B) = \frac{4}{5} \times \frac{3}{4} = \frac{12}{20} = \frac{3}{5}$

App III (2) None will hit = ? $\Rightarrow P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B}) = \frac{1}{5} \times \frac{1}{4} = \frac{1}{20}$

App III (3) P(at least one will hit) = ? $= 1 - P(\text{None will hit}) = 1 - \frac{1}{20} = \frac{19}{20}$

App III (4) P(either A or B or Both will hit) = ? = Same as part (3)

M-II $P(A \cup B) = ? = P(A) + P(B) - P(A \cap B) = \frac{4}{5} + \frac{3}{4} - \frac{3}{5} = \frac{19}{20}$

(5) Appl III Find the prob that target will be hit = ? = P(at least one will hit) = $\frac{19}{20}$

(6) Appl III Find the prob that A hit & B missed = ? = $P(A \cap \bar{B}) = \frac{4}{5} \times \frac{1}{4} = \frac{1}{5}$

(7) Appl III " " " " only one will hit = ?

$$P[(A \cap \bar{B}) \cup (\bar{A} \cap B)] = P(E_2 \cup E_3)$$

($A \cap \bar{B}$ → ME ← $\bar{A} \cap B$)

$= P(E_2) + P(E_3)$ ($\because E_2$ & E_3 are ME)

$= P(A \cap \bar{B}) + P(\bar{A} \cap B)$

$= \frac{4}{5} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4}$ ($\because$ A & $\bar{B}$ are Ind, similarly $\bar{A}$ & B are Ind)

$= \frac{7}{20}$

(this is the perfect eg to understand how to use the concept of ME & Independency in a single Quest.)

(8) If only one person hit then find the prob that A hit & B missed = ?

Prob III

Condition

original Prob = $P(S) = 1$

Reduced Prob = $P(\text{Condition}) = P(\text{only one person hit}) = \frac{7}{20}$ (using part (7))

fav Prob = $P(A \text{ hit \& } B \text{ missed}) = P(A \cap \bar{B}) = \frac{1}{5} = \frac{1/5}{1}$ (using part 6)

And Prob = $\frac{\text{fav Prob}}{\text{Reduced Prob}} = \frac{1/5}{7/20} = \frac{4}{7}$

(P8) Two Dice are rolled simultaneously & sum of the Nos on upper faces is observed to be 6 then find the Prob that Number 4 has appeared at least once?

Sol: original S.Sp = $\{(11)(12)\ldots(16),(21)(22)\ldots(26),(31)(32)\ldots\ldots\ldots(66)\} \Rightarrow n(S) = 36$ pair

Reduced Cases = $\{(15),(51),(24),(42),(33),(06),(60)\}$ ie 5 pair (Sum is 6)

fav Cases = $\{$No 4 has appeared at least once$\} = \{(24),(42)\}$ ie 2 pair

$$\text{Conditional Prob} = \frac{\text{fav Cases}}{\text{R.S.Sp}} = \frac{2}{5}$$

Note: W/o Cond^n, Ans = ? = P(No 4 has appeared at least once) $= \frac{f}{T} = \frac{11}{36}$

Q (2015) Parcels are sending from sender S to Receiver R sequentially through two post offices. The prob of loosing an incoming parcel is $\frac{1}{5}$ by each post office. independently of all other parcels. Given that Parcel has lost then find the Prob that it was lost by 2nd P.O ?

Sol $P(L) = \frac{1}{5}$ & $P(NL) = \frac{4}{5}$

Let we have Total (25) Parcels

$P(L \text{ by } 1^{st} P.O) = \frac{1}{5} = \frac{5}{25}$ & $P(NL \text{ by } 1^{st} P.O) = \frac{20}{25}$

$P(L \text{ by } 2^{nd} P.O) = \frac{1}{5} = \frac{4}{20}$ & $P(NL \text{ by } 2^{nd} P.O) = \frac{16}{20}$

$S \xrightarrow[25]{} (P.O\,1) \xrightarrow{\frac{1}{5}} (P.O\,2) \xrightarrow{\frac{1}{5}} R$

$(=25)$ $(=20)$ $(=16)$

original Prob = P(S) = 1

Reduced Prob = P(condition) = P(parcel is lost)

$$= P[\text{either lost by } 1^{st} \text{ or lost by } 2^{nd}]$$

(ME)

$$= P(\text{lost by } 1^{st}) + P(\text{lost by } 2^{nd})$$

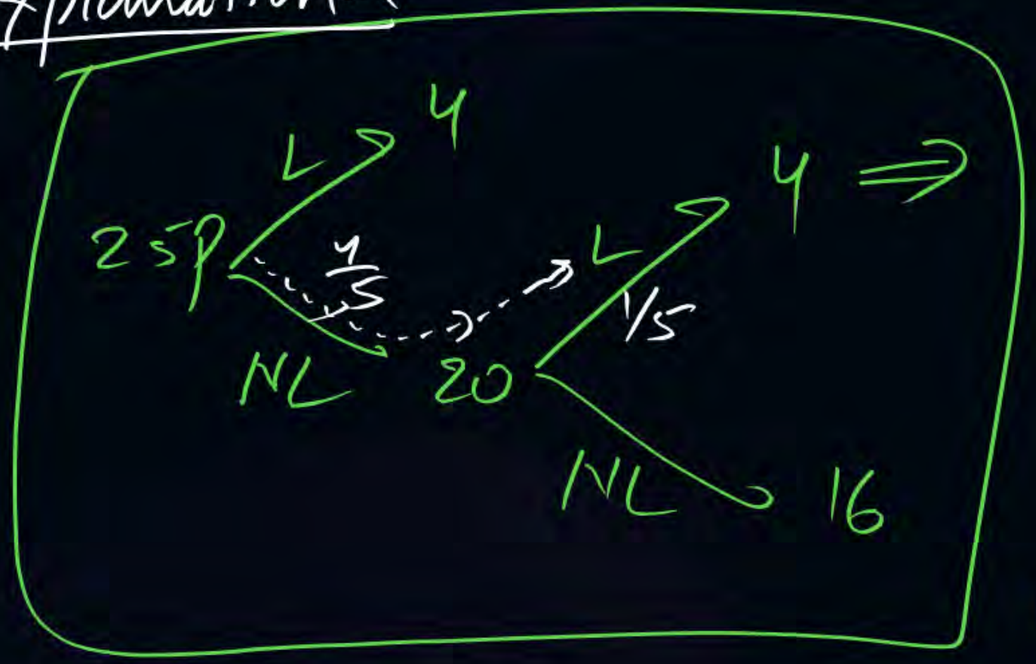
$$= P(\text{lost by } 1^{st}) + P[(\text{NL by } 1^{st}) \,\&\, (\text{Lost by } 2^{nd})]$$

$$= \frac{1}{5} + \frac{4}{5} \times \frac{1}{5} = \frac{9}{25}$$

fav Prob = P(Lost by 2nd P.O) = $P[\text{NL by } 1^{st} \,\&\, \text{Lost by } 2^{nd}] = \frac{4}{5} \times \frac{1}{5} = \frac{4}{25} = \frac{4/25}{1}$

So Conditional Prob = $\frac{\text{fav Prob}}{\text{R. Prob}} = \frac{4/25}{9/25} = \frac{4}{9}$

Explanation :-



$P(\text{Lost by } 2^{nd}) = P(\text{NL by } 1^{st} \text{ \& L by } 2^{nd})$

$= \frac{4}{5} \times \frac{1}{5} = \frac{4}{25}$

A single parcel can't be loose by (Both) the P.O at a time

$\Rightarrow$ ME Event

PO1: $S_1 = \{L, NL\}$

PO2: $S_2 = \{L, NL\}$

M-II Using Bayes' Th. → will be discussed later.

Standard Results of Conditional Probability :-

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = P[A \text{ when } B \text{ has already occurred}]$$

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = P[B \text{ when } A \text{ " " "}]$$

$$P[A \cap B/C] = \frac{P(A \cap B \cap C)}{P(C)} = P[\text{Simultaneous occurrence of } A \text{ \& } B \text{ when } C \text{ has already occurred}]$$

Multi Th.e $\Rightarrow$ $P(A \cap B) = P(A/B) \cdot P(B)$

Note:- (1) "In case of Independent Events, condition has no Meaning"

for eg Let A & B are Ind then

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{P(B) \cdot P(A)}{P(A)} = P(B)$$

(2) To check Independency of Events, we have following three methods;

M-I By using Def^{n}.

M-II if $P(A \cap B) = P(A) \cdot P(B)$ $\Leftrightarrow$ then [A & B are Ind.]

M-III if $P(A/B) = P(A)$ $\Leftrightarrow$ then [A & B are Ind.]

Q If $P(A)=\frac{1}{2}$, $P(B)=\frac{1}{3}$, $P(A/B)=\frac{1}{6}$

Then find the prob of their simultaneous occurrence?

$P(A \cap B)=?$

Sol:

$\because P(A/B) \neq P(A) \Rightarrow A$ & B are (Not Ind.) By M-III

is we can't Multiply directly

So we will use Multiplication Theorem:

$$P(A \cap B)=P(A/B).P(B)$$

$$=\frac{1}{6}\times\frac{1}{3}=\frac{1}{18}$$

Q If $P(A)=1$, $P(B)=\frac{1}{2}$

Then $P(A/B)=?$ & $P(B/A)=?$

$\because P(A)=1$

$\Downarrow$

A is sure Event

$\Downarrow$

A will definitely occur

$\Downarrow$

A is Ind from B (By M-I)

$\Downarrow$

B is also Ind from A

Hence $P(A/B)=P(A)=1$

$P(B/A)=P(B)=1/2$

⊛ the Concept of Indefendency as well as Depedency follow symmetric Relation (vice-versa)

ie A is Ind from B $\iff$ B is Ind from A

& A is Dependent on B $\iff$ B is also Dep on A

⊛ Doubt of Student :-

$$P(A \cap B) = P(A/B) \cdot P(B)$$

$= ?$

&

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$= ?$

Q. An hydraulic structure has four Gates, which operates independently. The prob of failure of each Gate is 0.2. Given that Gate 1 has failed then find the prob that Gate 2 and Gate 3 will also fail?

Sol: $P(G_1)=P(G_2)=P(G_3)=P(G_4)=0.2$

G_1, G_2, G_3, G_4 operates Independently.

$$P\left[\frac{G_2 \cap G_3}{G_1}\right] = ? = P[G_2 \cap G_3]$$

$$= 0.2 \times 0.2 = 0.04$$

M-II

$$\text{Req Prob} = P[G_1 \cap G_2 \cap G_3]$$

$$= 1 \times 0.2 \times 0.2$$

$$= 0.04$$

Qe A coin is tossed twice & Consider two Events A & B as follows;

$A = \{$ Both H & T appeared $\}$, $B = \{$ At Most one T in appeared $\}$

then evaluate $P(A), P(B), P(A/B), P(B/A) = ?$

Sol: $S = \{(HH), (HT), (TH), (TT)\} \Rightarrow n(S) = 4$

$A = \{(HT), (TH)\} \Rightarrow P(A) = \frac{2}{4} = \frac{1}{2}$

$B = \{(HH), (HT), (TH)\} \Rightarrow P(B) = \frac{3}{4}$

$A \cap B = \{(HT), (TH)\} \Rightarrow P(A \cap B) = \frac{2}{4} = \frac{1}{2}$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/2}{3/4} = \frac{2}{3}$$

$$P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{1/2}{1/2} = 1$$

Qe Two persons p & q are considering to apply for job.

The prob that p apply for a job is $\frac{1}{4}$.

The prob that p apply for a job given that q apply for a job is $\frac{1}{2}$

& " " q " " " " given that p " " " " is $\frac{1}{3}$ then find

" " p does not apply for a job given that q does not apply for a job ?

Sol: $P(p)=\frac{1}{4}$, $P(p/q)=\frac{1}{2}$ & $P(\bar{p}/\bar{q})=?$

$P(q)=?$, $P(q/p)=\frac{1}{3}$

p $\bar{p}$ S_1 , q $\bar{q}$ S_2 $\Rightarrow$ $S=\{p/q, p/\bar{q}, \bar{p}/q, \bar{p}/\bar{q}, q/p, q/\bar{p}, \bar{q}/p, \bar{q}/\bar{p}\}$

ATQ, $P(a/p) = \frac{1}{3}$

$$\frac{P(a \cap p)}{P(p)} = \frac{1}{3}$$

$$\Rightarrow P(a \cap p) = \frac{1}{3} P(p) = \frac{1}{3} \times \frac{1}{4} \left(\frac{1}{12}\right)$$

again ATQ,

$$P(p/a) = \frac{1}{2}$$

$$\frac{P(p \cap a)}{P(a)} = \frac{1}{2}$$

$$P(a) = 2P(p \cap a) = 2 \times \frac{1}{12} \left(\frac{1}{6}\right)$$

$$P(\bar{p}/\bar{a}) = \frac{P(\bar{p} \cap \bar{a})}{P(\bar{a})} = \frac{1 - P(p \cup a)}{1 - P(a)}$$

$$= \frac{1 - \{P(p) + P(a) - P(p \cap a)\}}{1 - P(a)}$$

$$= \frac{1 - \left\{\frac{1}{4} + \frac{1}{6} - \frac{1}{12}\right\}}{1 - \frac{1}{6}} = \left(\frac{4}{5}\right) \text{ Ans}$$

Note: $\because P(p) = \frac{1}{4}$ then $P(a) = ? = 1 - P(p) = 1 - \frac{1}{4} = \frac{3}{4}$

what is wrong in it ??

$\because p$ & a are Ind & $P(p) + P(a) \neq 1$

NWB Consider an unbalanced Die (numbered 1,2,3,4,5,6).
the prob of an odd face is 90% Prob of an Even face,
& Prob of any Even numbered face is Same.
The Prob of Even face given that face value exceed 3 is 0.75
then find the prob that Face Value exceeds 3?

Soln: $P(\text{odd f}) = 90\%\ P(\text{Even f})$

i.e $P(\text{odd f}) = \frac{9}{10} P(\text{Even f})$

Die $= \{1,2,3,4,5,6\}$
i.e Each outcome is ME

again, $P(2) = P(4) = P(6) = n$ (let)

$P(\text{Even f} / f>3) = 0.75$ then $P(f>3) = ?$

W.K.T,

$P(1)+P(2)+P(3)+P(4)+P(5)+P(6)=1$

is $P(\text{odd } f)+P(\text{Even } f)=1$

$\frac{9}{10}P(\text{even } f)+P(\text{even } f)=1$

$P(\text{even } f)\left[\frac{9}{10}+1\right]=1$

ie $P(\text{even } f)=\frac{10}{19}$

or $P(2)+P(4)+P(6)=\frac{10}{19}$

$x+x+x=\frac{10}{19} \Rightarrow x=\boxed{\frac{10}{57}}$

$P(\text{Even } f/f>3)=0.75$

$$\frac{P[\text{Even } f \cap f>3]}{P[f>3]}=\frac{3}{4}$$

$$\frac{P[4 \text{ or } 6]}{P(f>3)}=\frac{3}{4}$$

$$\frac{P(4)+P(6)}{P(f>3)}=\frac{3}{4}$$

$$\frac{x+x}{P(f>3)}=\frac{3}{4}$$

$P(f>3)=\frac{3x}{3}$

$=\frac{3\times\left(\frac{10}{57}\right)}{3}$

$=\boxed{\frac{80}{171}}$

$=0.467$

H.Q.2 A coin is tossed twice then $P(\text{Both are H}) = ?$ if we have at least one Head

Ans $= \frac{1}{3}$

THANK - YOU